Selection and screening of stable mammalian cell lines

Risk assessment

– Work with human-derived material or transgenic cell lines (BSL-2)
▷ Wear gloves, safety glasses, lab coat
◻ Collect and dispose waste after inactivation as REGULATED MEDICAL WASTE

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This is a bench card. Full protocol available online.

A › *Antibiotic selection of mammalian cell lines*

◻ Culture medium with antibiotics

(1.) Determine the minimum effective antibiotic concentration for selection using a matrix titration format.

- Prepare a 96-well plate with 100 µL of culture medium in all wells.
- Perform a row-wise dilution of the antibiotic: Add 100 µL of the highest concentration to row A, and perform 1:2 serial dilutions down to row G. Leave row H as a no-antibiotic control.
- In each row, seed column 1 with 100 µL of a $4.0{\times}10^4$ mL^{-1} cell suspension. Mix gently and transfer 100 µL to column 2. Continue serial dilution across columns to column 12. Discard excess.
- Incubate the plate under standard conditions.
- Assess cell survival under the microscope after 3–7 d.

(2.) Select cells using the lowest antibiotic concentration that resulted in complete killing of non-transgenic cells. Include a negative control (non-transfected) to verify selection stringency. Useful ranges:

Cell line	**Blasticidin S**	**G418 (Geneticin)**	**Hygromycin B**	**Puromycin**	**Zeocin**
Typical duration	**10 days**	**14 days**	**4 days**	**4 days**	**10 days**
293T	2–10 µg/mL	Not determined	Not determined	1.0–5.0 µg/mL	Not determined
HeLa	5–10 µg/mL	Not determined	Not determined	1.0–2.0 µg/mL	Not determined
MCF 10A	2–10 µg/mL	150–400 µg/mL	100 µg/mL	0.5–2.5 µg/mL	750 µg/mL
iPS	10–20 µg/mL	Not determined	Not determined	Not determined	Not determined

(3.) Maintain selection by feeding cells regularly. Expand once resistant colonies are visible.

(4.) *Optional:* Perform a second round of limiting dilution under selection to ensure the selected cell population is monoclonal.

B › *Screening of mammalian cell lines*

(1.) Physically separate the population into clones using limiting dilution, pipette-based cell picking, or flow cytometry (FACS).

(2.) Feed cells regularly and monitor for clonal outgrowth. Split as needed.

> ***Critical:*** In 96-well plates, border wells lose volume faster than center wells. Top up edge wells regularly to prevent desiccation.

(3.) Analyze each clone for expression of the transgene or phenotypic marker.

Recipe (available online) Resources (available online) Troubleshooting (available online) Notes (available online)